

The Prescriptive Imperative: Rethinking Mechanical Fault Reliability in EOT & Ladle Cranes

Why mechanical fault intelligence is redefining crane reliability in heavy industry and why the Crane ai shield solution captures on operating reality, not on a clock.



AUTHORS:

Chaitanya Bulusu, Senior Vice President and Head, Infinite Uptime

Priyanka Gupta, Category Manager - Process & Energy Efficiency, Infinite Uptime

EXECUTIVE SUMMARY



Heavy industry has spent a decade digitizing the plant floor. The unsolved problem is no longer data collection - it is converting continuous sensor data into decisions a maintenance team can act on before failure, not after a probability score fires.



Cranes are among the highest-consequence rotating assets in most plants, and the monitoring approaches built for steady-state pumps and motors structurally fail on crane duty cycles - sampling gaps, load-blind sensors, and environmental sensor failure compound into a 2-6-week blind spot before catastrophic failure.



Crane AI Shield closes that gap as a solution - the PlantOS™ product and Infinite Uptime's service offering together, delivering a defined service-level agreement (SLA). It captures on operating reality rather than on a clock (24x7x365, RPM-gated, always synchronized to load), through a purpose-built three-tier architecture: the Sensing Layer, the Edge



Reliability Engine, and the Prescriptive AI Engine Layer - with Infinite Uptime's Center of Excellence operating in parallel to Tier 3. What plants buy is a maintained reliability outcome - sensors are the enabling technology, the service keeps the plant at the highest SLA levels.



A defining element of the solution is the Infinite Uptime Center of Excellence (COE). Every prescription the AI engine produces is validated by CAT III and CAT IV certified reliability experts before it is released to the plant. The COE operates 24x7x365 in parallel with Tier 3, placing certified reliability specialists between the AI engine and the maintenance team.



Two steel-industry deployments - one fleet-scale and one single-crane early-warning catch - demonstrate both portfolio-level reliability impact and asset-level failure prevention in practice.



Adoption requirements are limited: no crane modifications, no equipment replacement, one planned shutdown window, and go-live in 5 days. Reported payback is 6-12 months.